

Database Management System

(Notes)

Part = 3

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The Relational Data model and Relational Database constraints.

Relational model concepts:

→ The relational model is the most widely used data model in modern database management system (DBMS).

→ It was proposed by Edgar. F Codd in 1970.

→ In the relational model data is stored in the form of tables called relations. Each table consists of rows and columns.

Example

Student database

Student Id	Name	Department	Age
101	Ram	CSIT	21
102	Sita	BIT	22
103	Hari	BCA	22

↳ The entire table is a Relation. Student.

↳ Rows represent individual records.

↳ Columns represent student's properties.

Basic Terminologies of Relational Model:

① Relation

→ A Relation is a table containing rows & columns

Example Student

Student-ID	Name	Age
101	Ram	21
102	Sita	20

→ The entire table is called a Relation.

② Tuple

→ A Tuple is a single row of a relation.

Example: 101, Ram, 21 → Tuple 1
102, Sita, 20 → Tuple 2

③ Attribute

→ An attribute is a column in a relation.

Example

Student, Attributes → Student-ID
Name
Age.

④ Domain

→ A Domain is the set of all possible valid values that an attribute can take

Example:

Attribute → Age,

Possible values → 18

19

22

60

Invalid value → Apple

-208

-15

-100

↳ So the domain defines what values are allowed.

⑤ Degree

→ The degree of a relation is the no. of attributes (columns).

Example. ID | Name | Age | Address |

↳ column → 4

Degree = 4

⑥ cardinality:

→ The cardinality of a relation is the number of tuples (rows)

Example

ID	Name	
1	Ram	→ Row 1
2	Sita	→ Row 2
3	Hari	→ Row 3

Rows = 3
cardinality = 3

⑦ Relation Schema

→ A Relation schema defines the structure of a relation.

Example: Student (Student-ID, name, Age, Department)

→ It tells → relation name

→ attributes name

but not the actual data.

⑧ Relation Instance
→ A relation Instance is the actual data stored in the relation at a particular time.

Schema : Student (ID, Name, Age)

Instance

ID	Name	Age
101	Ram	21
102	Sita	20

Properties (characteristics) of a Relation:

① Relation has a unique name.

② Attribute name are unique

Correct → ID / Name / Age ✓
Incorrect → ID / Name / name X

③ Order of rows does not matter.

Table 1 →

1 Ram

2 Sita

Table 2 →

2 Sita

1 Ram

→ Both are identical logically

(4) Order of columns does not matter.

(5) Every cell contains a single (atomic) value.

(6) Duplicate tuples are not allowed

e.g.

ID

name

1

Ram

→ X

1

Ram

(7) All values in a column come from the same domain.

eg.

ID

name

Age

(18-60)

101

A

18

102

B

20

103

C

22

104

D

35

105

E

60

Relational model notation:

(1) Relation (Table)

→ Represented by a capital letter or relation name,

e.g. STUDENT.

- ② Relation Schema
↳ Relation - name (Attribute 1, Attribute 2, Attribute N)
Example,

~~Student~~

STUDENT (ID, Name, Age)

- ③ Tuple (Row)
↳ Represent by a single word in a relation
Example:

(101, Ram, 21)

- ④ Attribute (column)
↳ Attribute are written inside the relation Schema

Example

... (ID, Name, Age)

- ⑤ Domain

↳ Denoted as $\text{dom}(\text{Attribute})$, representing the set of valid values for an attribute

Example

$\text{dom}(\text{Age}) = \{18, 19, 20, \dots, 100\}$

Categories of Constraints:

→ Constraints are rules enforced on a database to maintain the accuracy, consistency, and integrity of data. In the relational model, the three main categories of constraints are:

- (1) Domains constraints
- (2) Key constraints
- (3) Referential integrity constraints

1 Domains constraints

→ A domain constraint specifies the set of valid values that an attribute (column) can contain. Every value in a column must belong to its predefined domain.

Example: Student Table

Student

Student ID	Name	Age
101	Ram	21
102	Sita	20

SUPPOSE the domain of Age is 18-60

Valid value → 18, 19, 20 ... 60 ✓

Invalid val → 70, 80, -10, -20 ...

'tarento' ✗

→ These values violate the domain constraint because they are outside the allowed domain.

② Key constraints:

→ A key constraint ensures that every tuple (row) in a relation is uniquely identified.

→ No two rows can have the same value for the primary key.

Example

ID	Name	Age	
101	Ram	21	
102	Sita	20	✓
103	Hari	22	

ID → Primary key

Invalid Table:

ID	Name	Age	
101	Ram	21	✗
101	Sita	22	

③ Referential Integrity Constraints:

→ A referential integrity constraint ensures that a foreign key value in one table must either match an existing primary key value in the reference table or be null (if null values are allowed).

Entity

Entity Integrity, Referential Integrity & Foreign Key:

Entity Integrity:

→ Entity Integrity states, The primary key of relation cannot be null.

→ Every row (tuple) in a table must have a unique primary key value so that each record can be uniquely identified.

Example

Valid Table

ID (PK)	Name	Age
101	Ram	21
102	Sita	20

Invalid Table

ID (PK)	Name	Age
NULL	Ram	21
102	Sita	20

Concept of Insert, update and delete:

- ① Insert operation:
- ② Update operation:
- ③ Delete operation:

Dealing with Constraint Violation:

① Domain constraint violation,

↳ Occurs when an inserted or updated value does not belong to the defined domain of an attribute.

Example

Valid Age range is 18-60.

ID	Name	Age
101	Ram	65

This is invalid because 65 is outside the given domain.

Action: DBMS rejects the insert or update operation.

② Key Constraint violation

Action: DBMS rejects the operation.

③ Entity Integrity violation.

④ Referential Integrity violation